Quantum Leadership Training Program 2025

Quantum Machine Learning



Nov 24, 2025

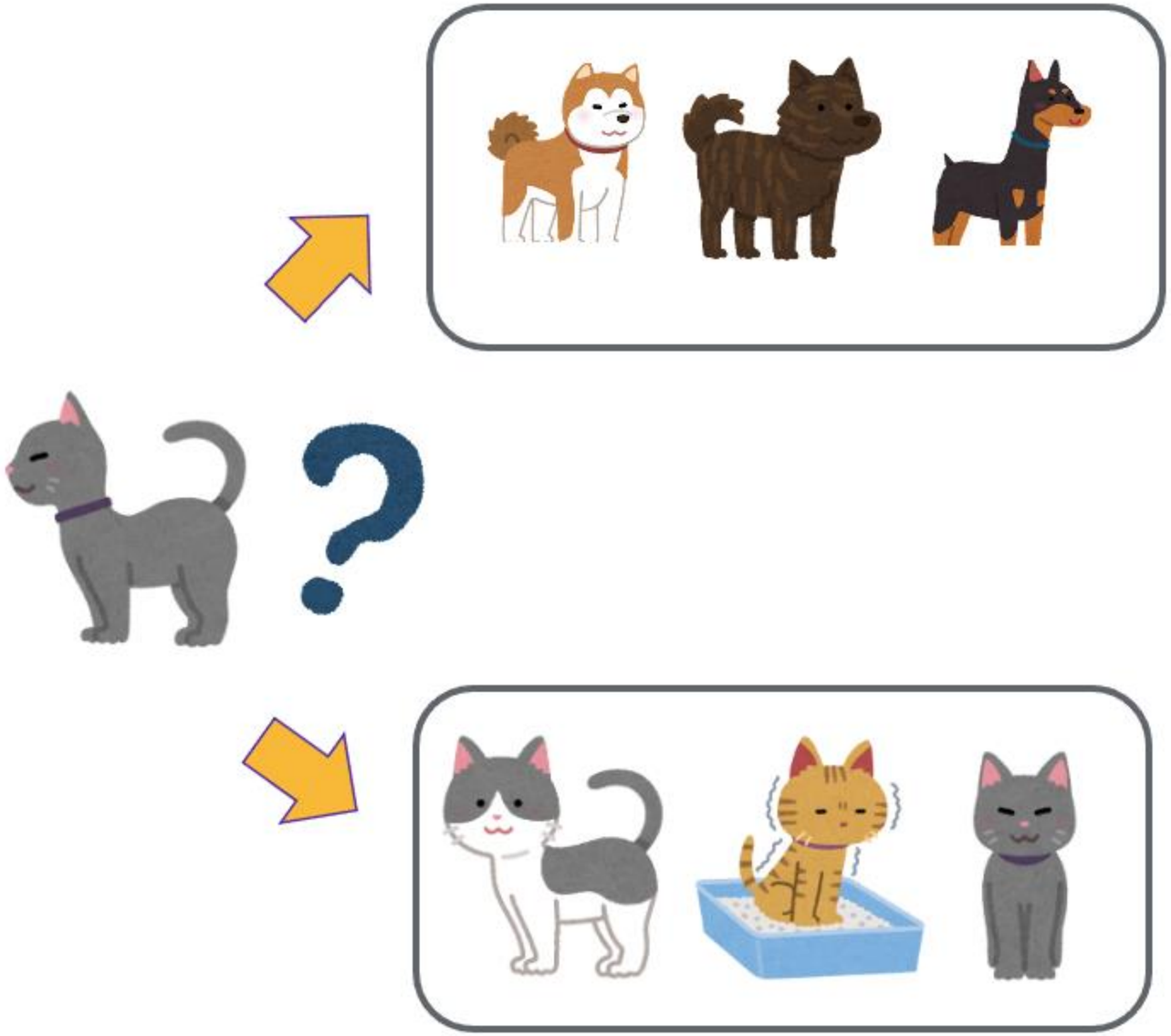
Kifumi Numata

IBM Quantum

kifumi@jp.ibm.com

IBM **Quantum**

© 2025 International Business Machines Corporation



NETFLIX

Home Shows Movies Games New & Popular My List Browse by Languages

Kids 4

Your Next Watch











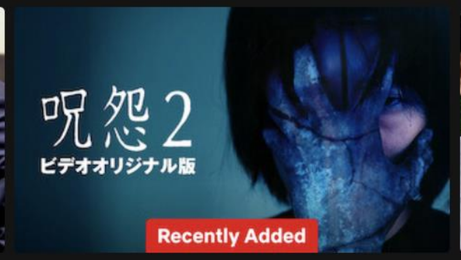


We Think You'll Love These














AI

OpenAI

Log in

What can I help with?

Teach me to negotiate

Learn about ChatGPT Business

Search with ChatGPT

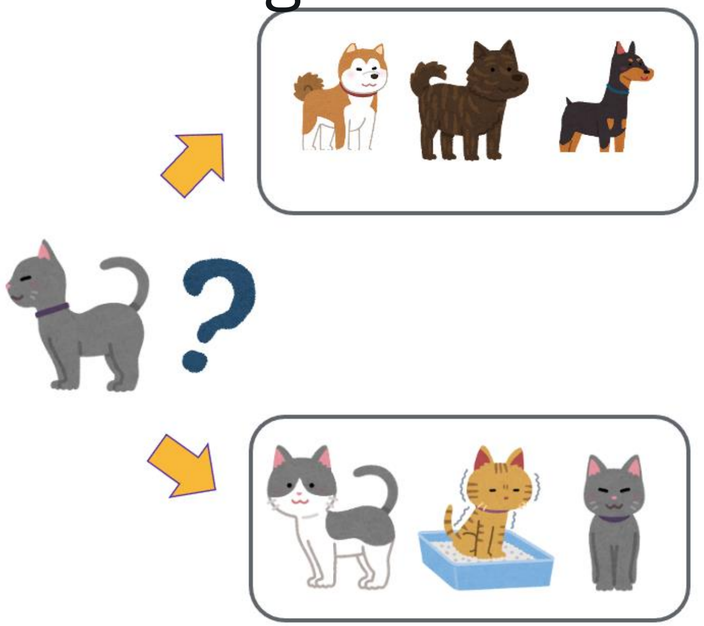
Talk with ChatGPT

Research

More

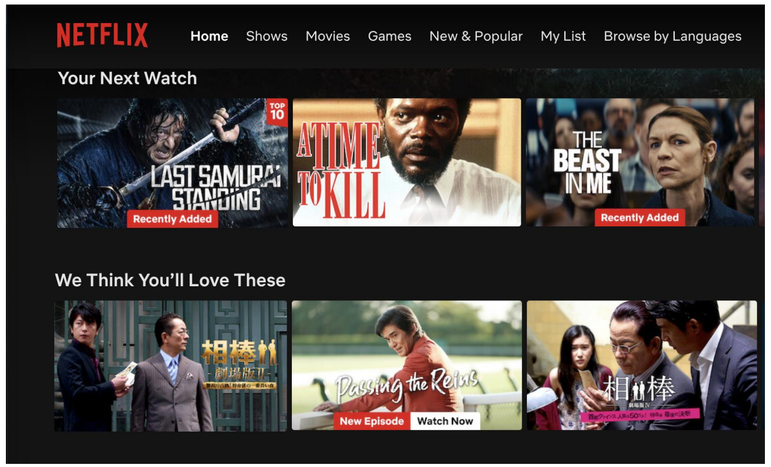
Types of machine learning

Supervised learning



Identify new photos of cats and dogs from a labeled collection of photos of cats and dogs.

Unsupervised learning



Group viewers based on their movie viewing history and recommend new movies to them.

Reinforcement learning

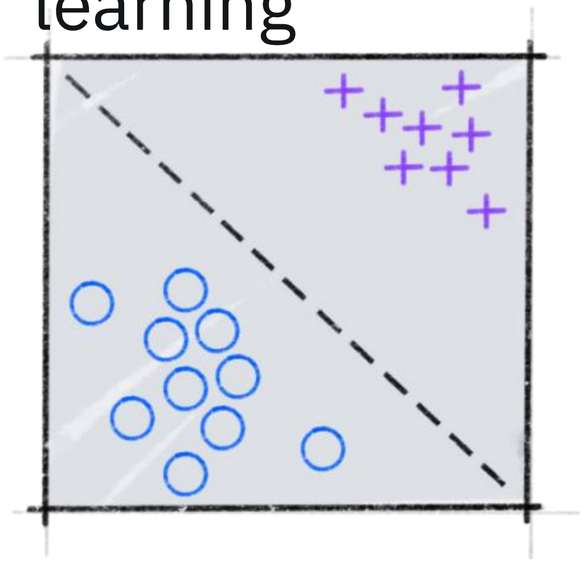


Learning how to play Go algorithmically.

Generative AI learns by combining these.

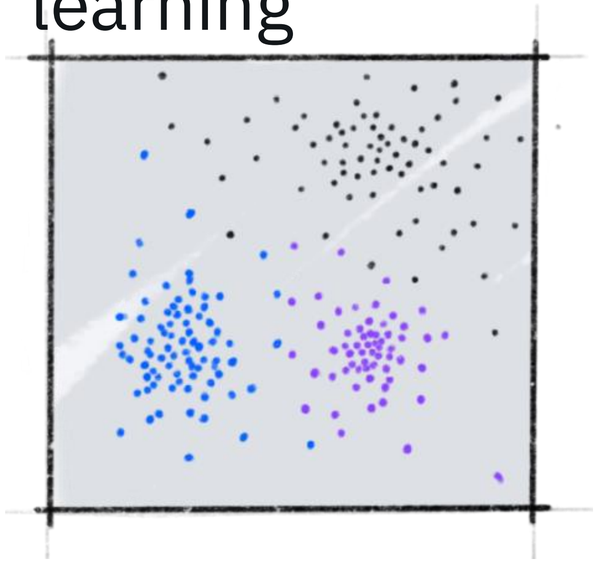
Types of machine learning

Supervised learning



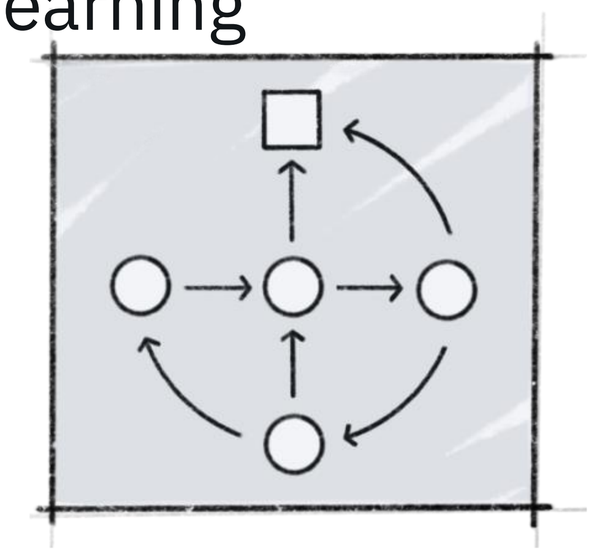
Labeled data (x_i, y_i):
Learn the function $y = f(x)$.

Unsupervised learning



Unlabeled data:
Discover hidden patterns and structure in data.

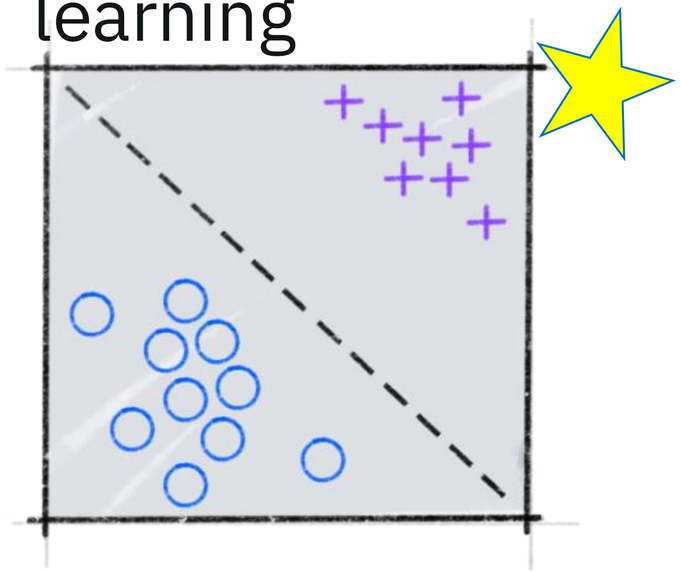
Reinforcement learning



Maximize expected rewards in a behavior-based reward environment.

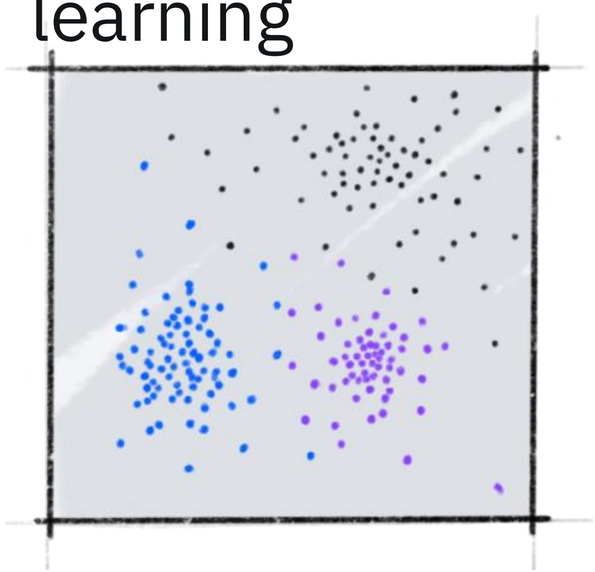
Types of machine learning

Supervised learning



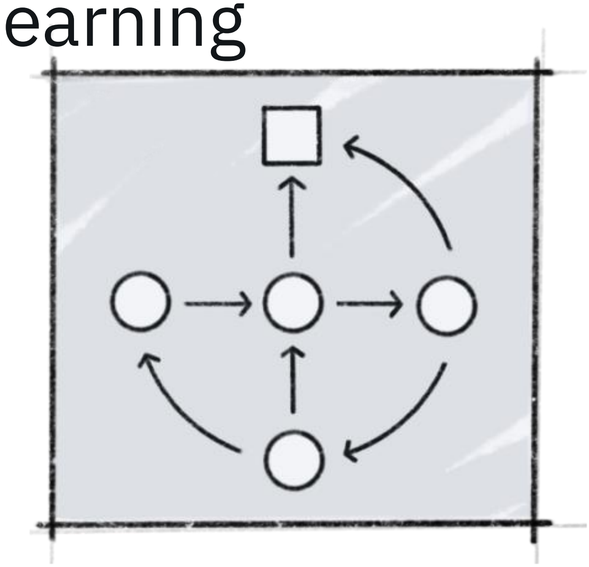
Labeled data (x_i, y_i):
Learn the function $y = f(x)$.

Unsupervised learning



Unlabeled data:
Discover hidden patterns and structure in data

Reinforcement learning



Maximize expected rewards in a behavior-based reward environment.

How does a computer distinguish between strawberries and apples?





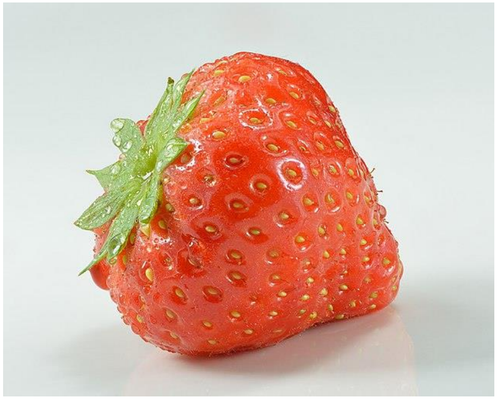










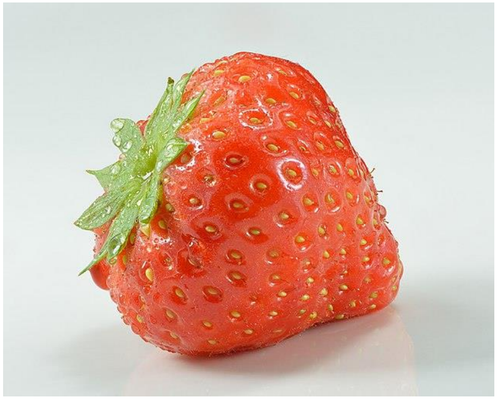


It identifies them based on their features.
Then, what are the features that distinguish strawberries from apples?







Strawberries and apples

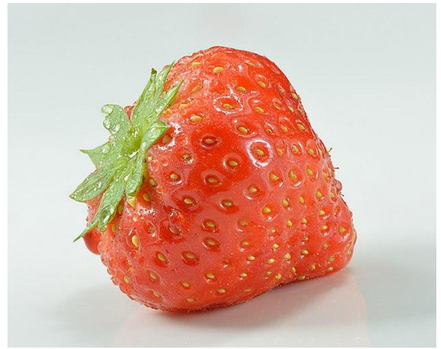


Round

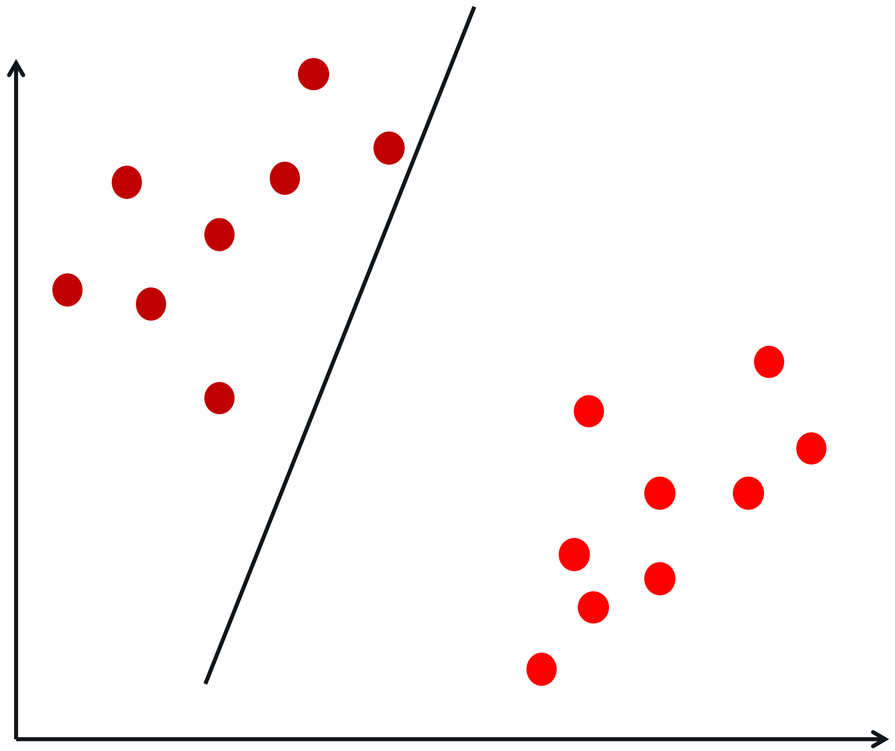
Sharp

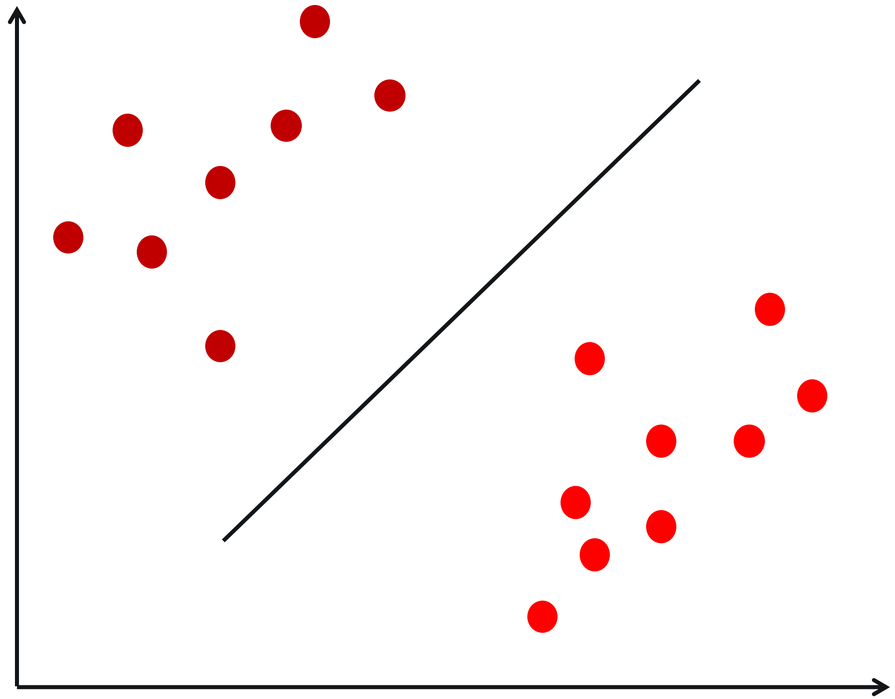
Smooth

Bumpy

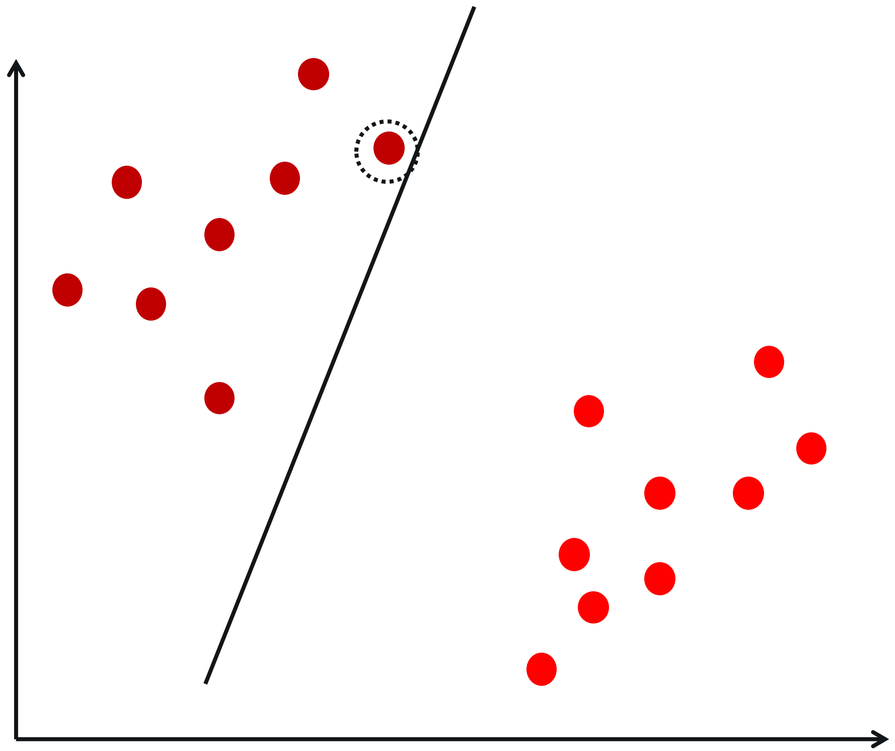


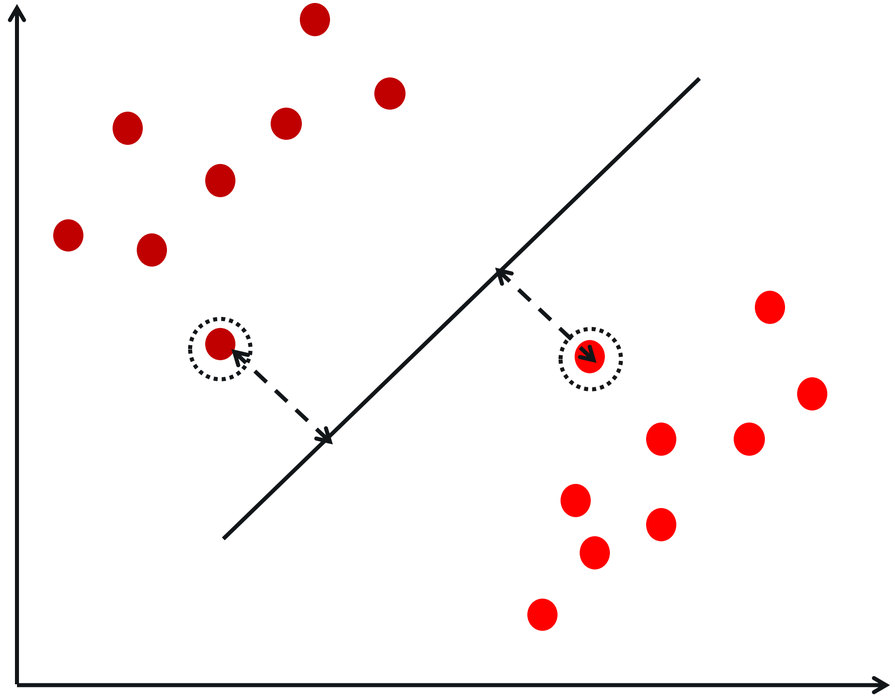
Which is better classified?





The distance between the boundary line and the nearest data point is longer in the right figure.



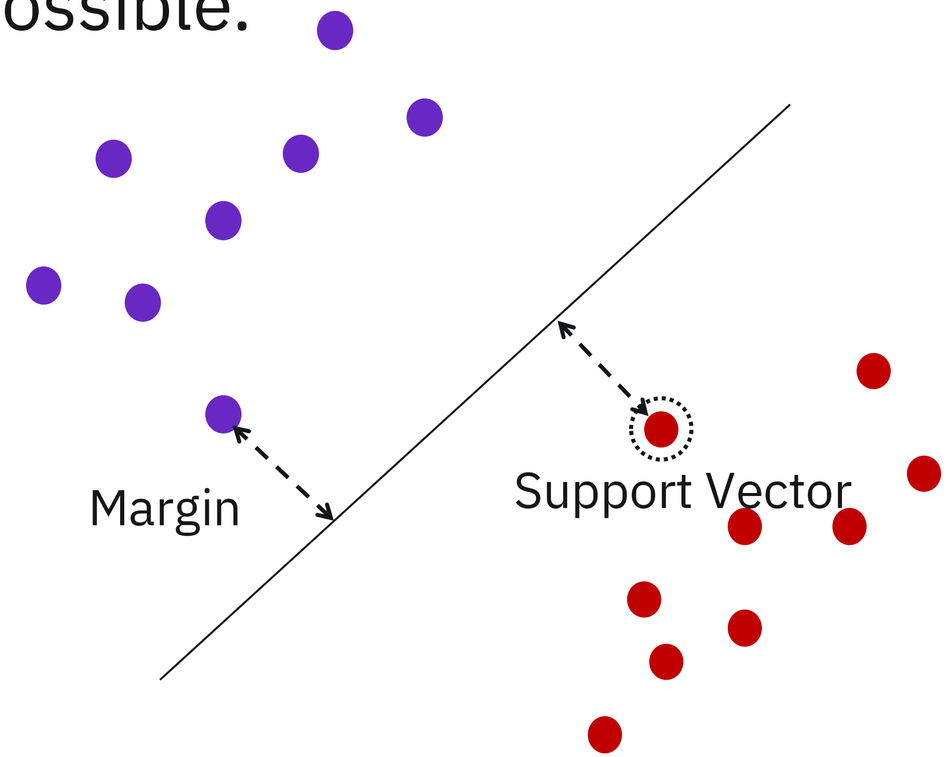


Better classified!

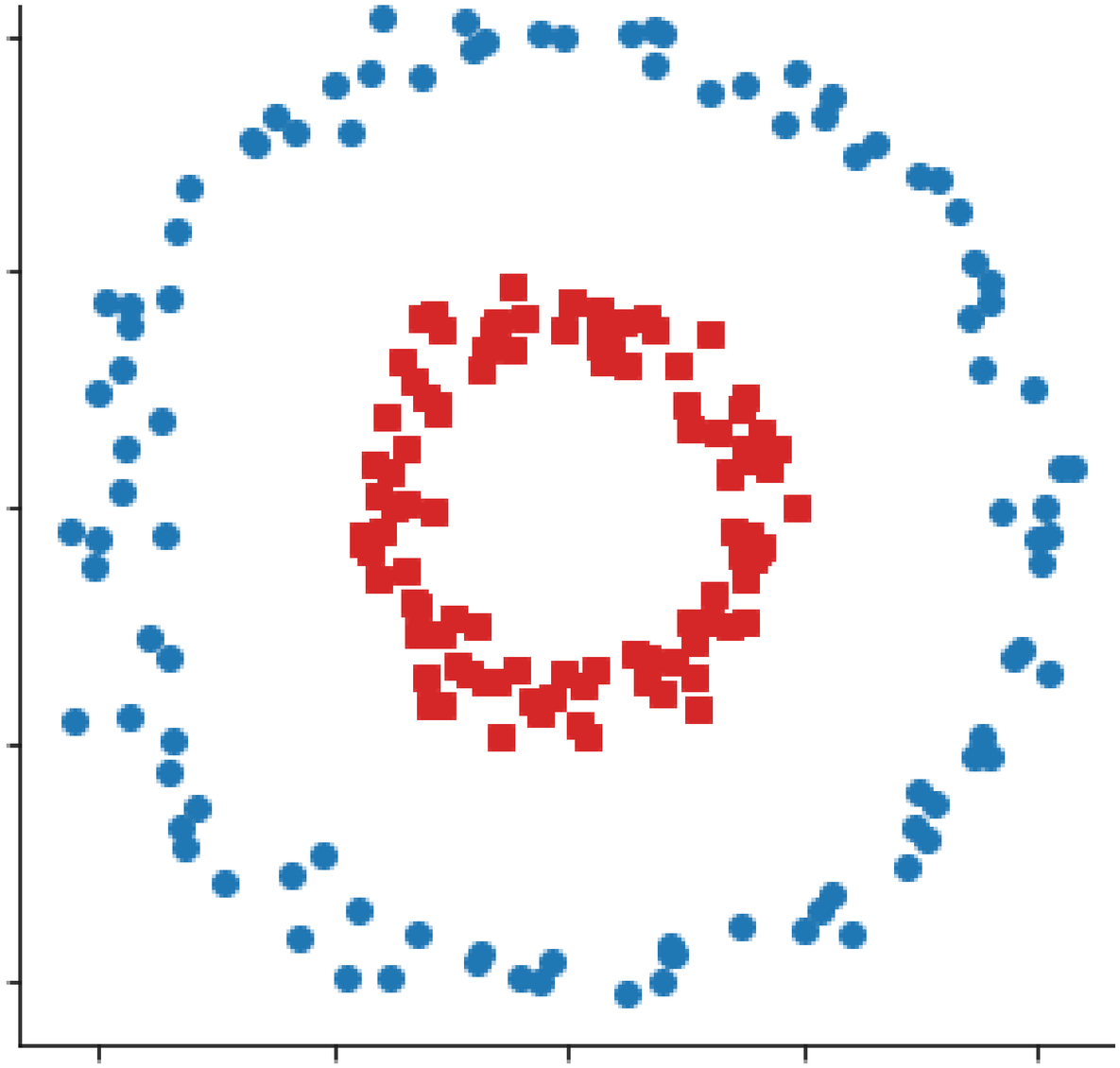
SVM (Support Vector Machine)

A method to divide data into two groups (binary classification)

- Analytical method to define boundary line between groups.
- Optimize margins, distance between boundary and nearest data point, to be as large as possible.



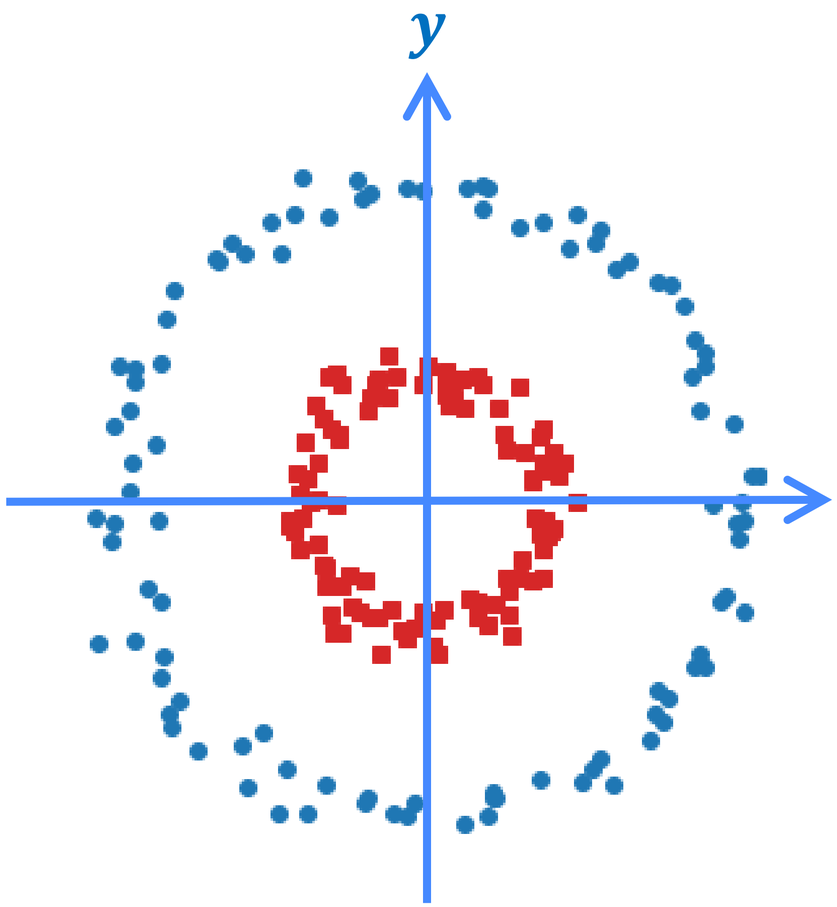
For data that cannot be divided by a line



It is clear that this feature set can be divided into two groups, but the boundaries are not linear.

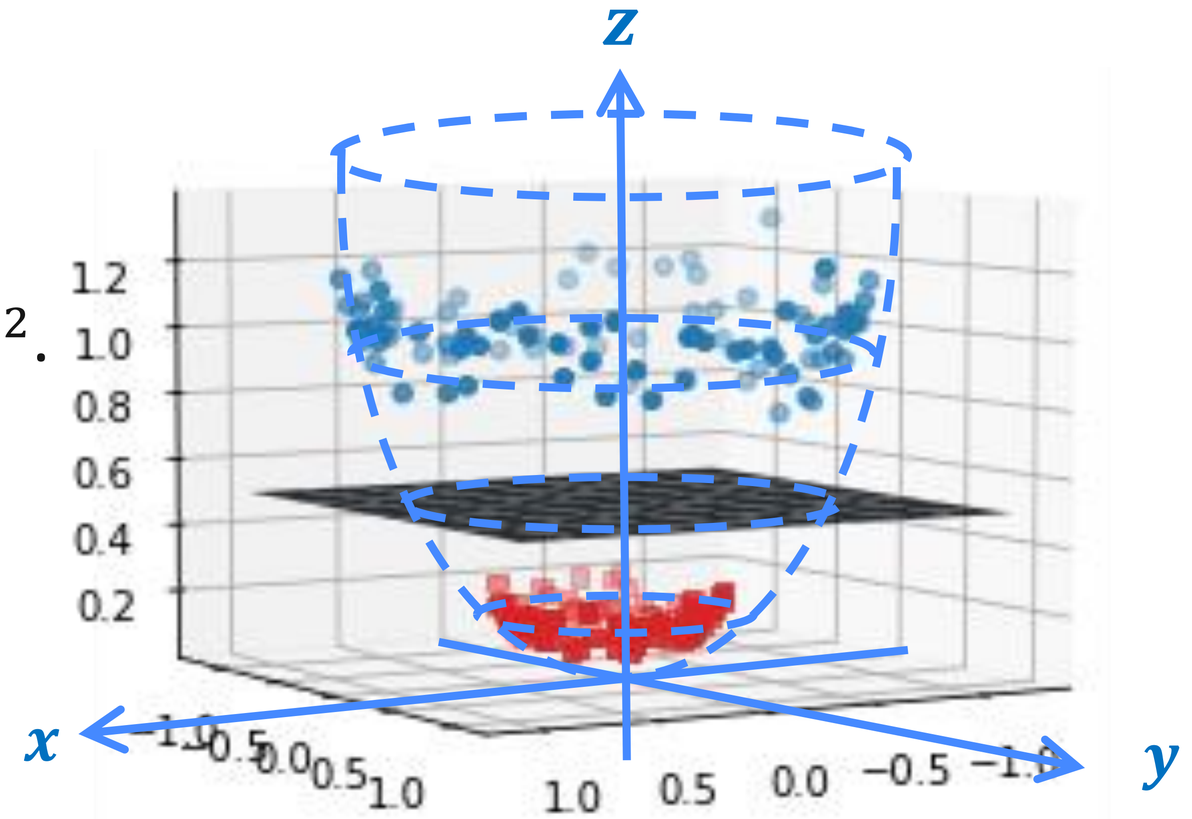
15

Feature mapping



Convert data to three dimension using $z = x^2 + y^2$.

Data can be separated by planes in three dimension.

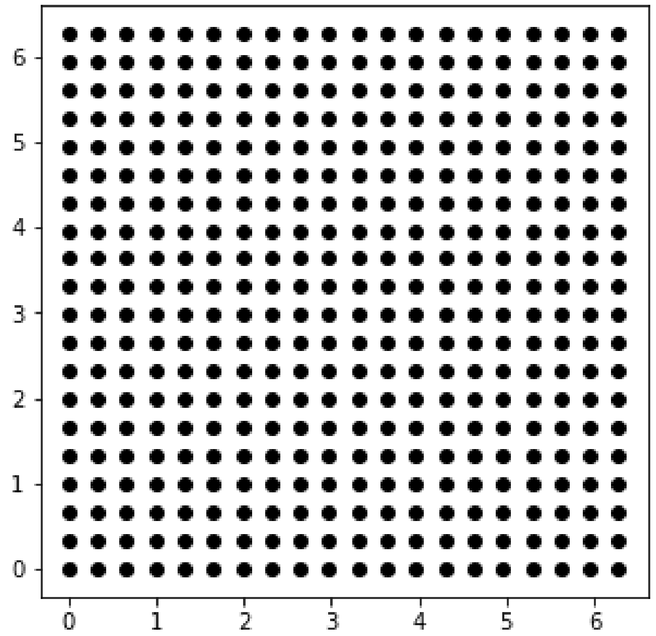


By increasing the dimensionality of the features (feature mapping), it is possible to separate them at a flat (linear) boundary surface.

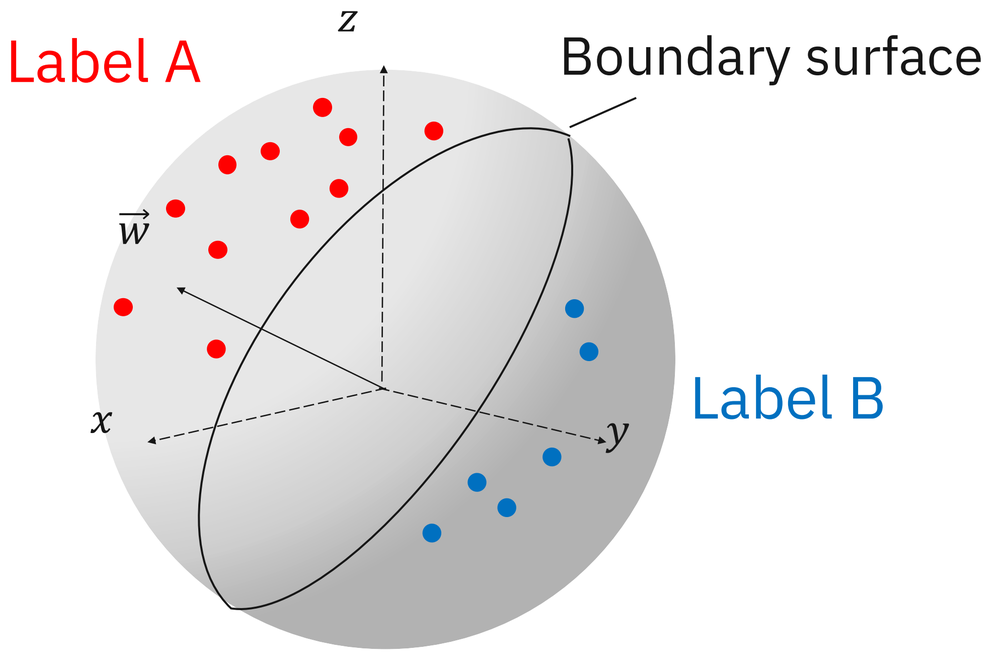
16

Feature mapping to quantum state

Mapping from low-dimensional feature sets to higher dimensional spaces allows data to be effectively separated and classified.



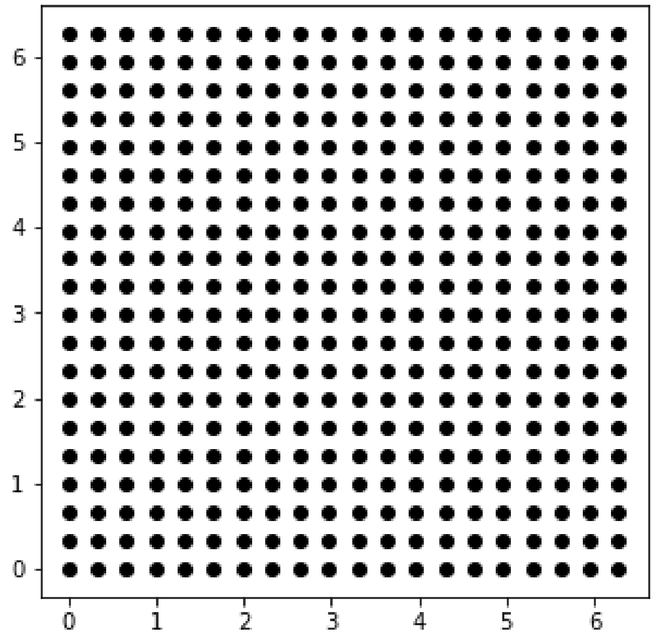
Map to quantum state



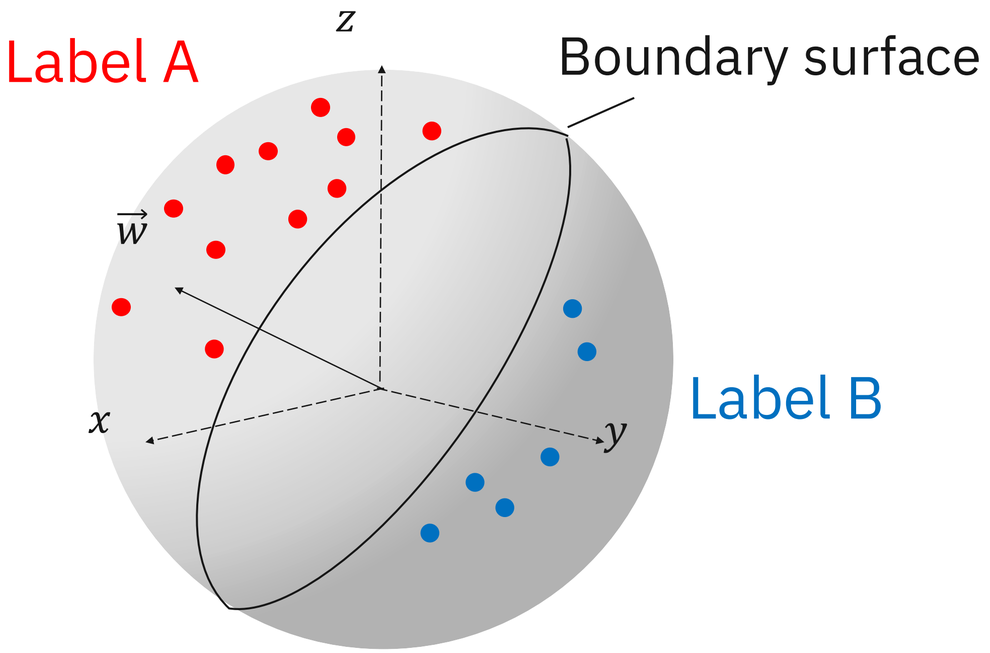
17

Feature mapping to quantum state (again)

Mapping from low-dimensional feature sets to higher dimensional spaces allows data to be effectively separated and classified.



Map to quantum state

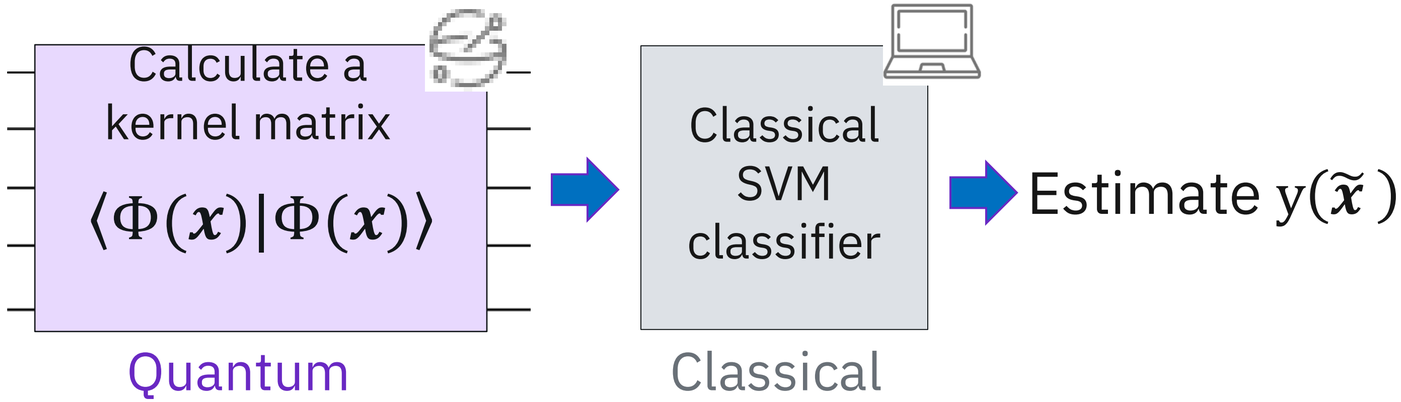


For creating the boundary surface, we will use classical SVM.

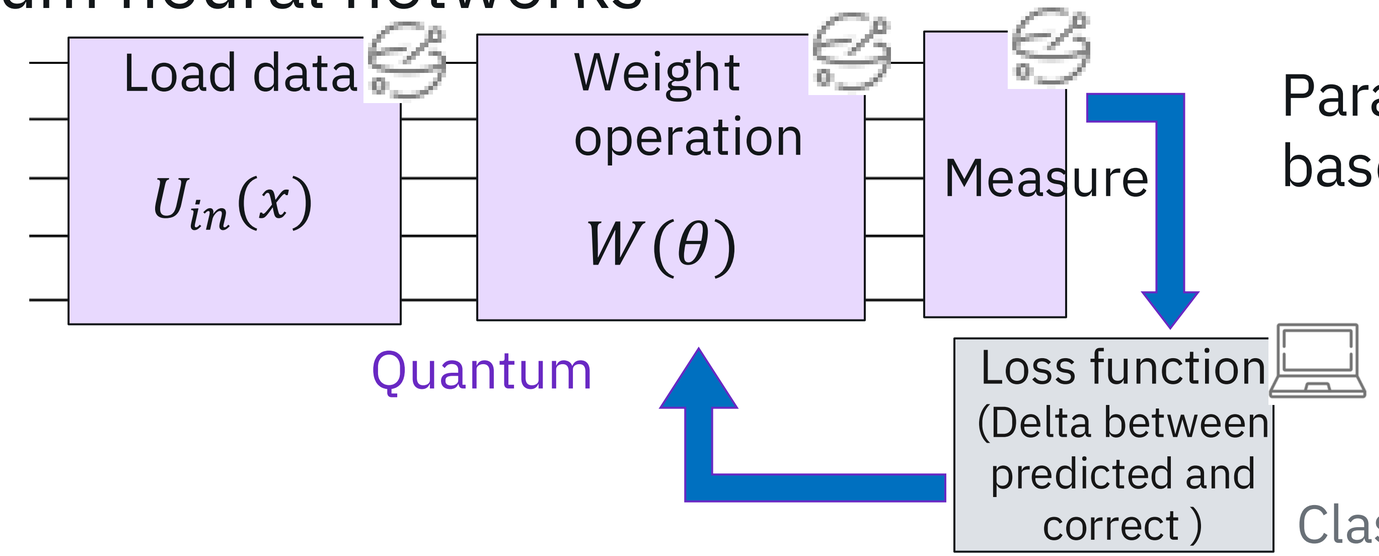
18

Two main quantum machine learning in supervised learning

Quantum Kernel estimation



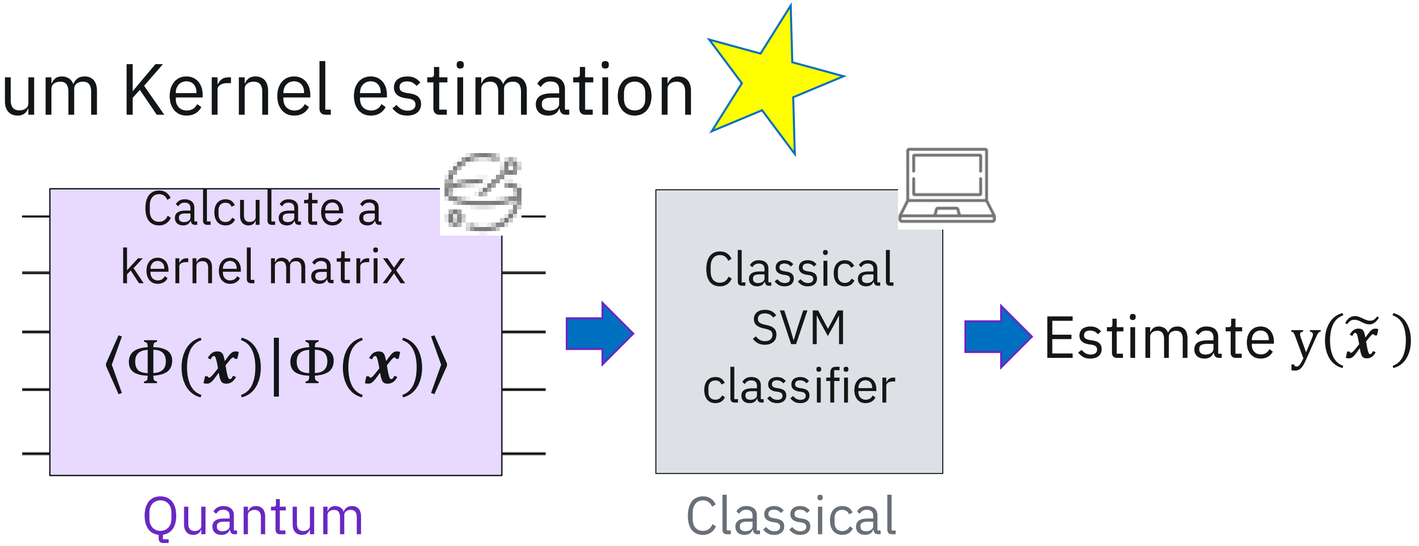
Quantum neural networks



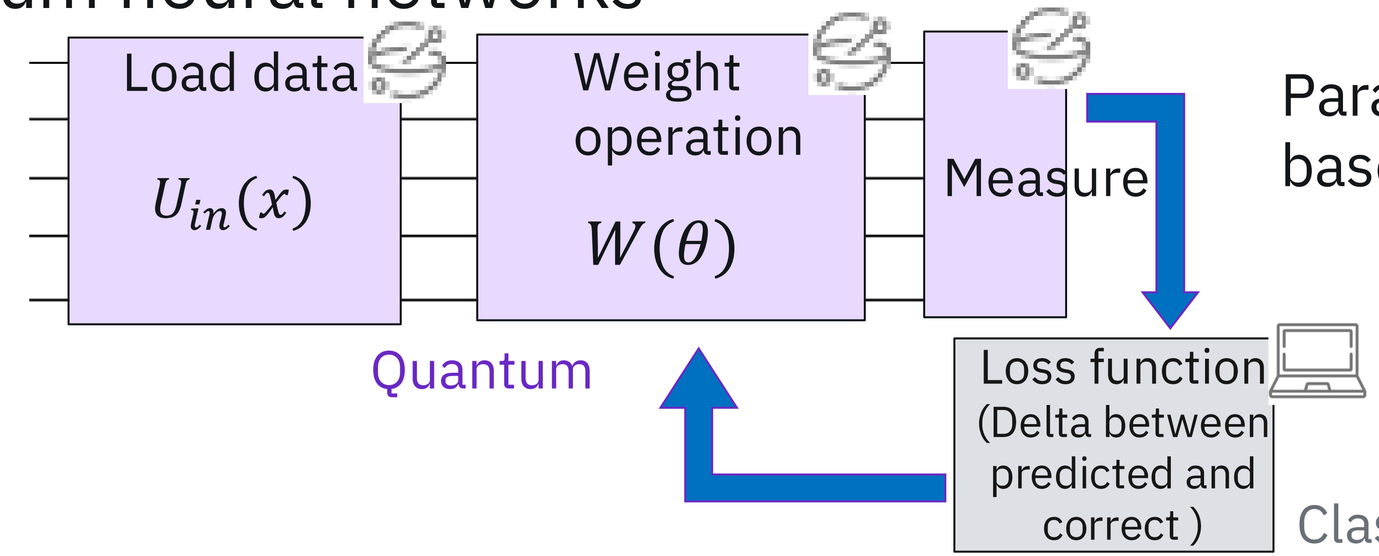
Parameter θ is optimized based on supervised data.

Two main quantum machine learning in supervised learning

Quantum Kernel estimation



Quantum neural networks



Parameter θ is optimized based on supervised data.

Reference:
https://quantum.cloud.ibm.com/learning/en/courses/quantum-machine-learning/qvc-qnn

Managing expectations

It is not true that some quantum machine learning algorithms can solve all, or even many, classification tasks more efficiently and scalably than classical machine learning algorithms.



We will study quantum machine learning to find the answer to below questions:

1. What quantum circuits are most likely to behave in novel ways, compared to classical alternatives?

2. Is there real-world data whose properties are best suited to using such novel quantum circuits?

3. Could such quantum circuits be realized in near-future quantum computers?

Typical Data encoding methods for QML

1. Basis encoding

2. Amplitude encoding

3. Angle encoding

4. Enhanced angle encoding

Basis encoding

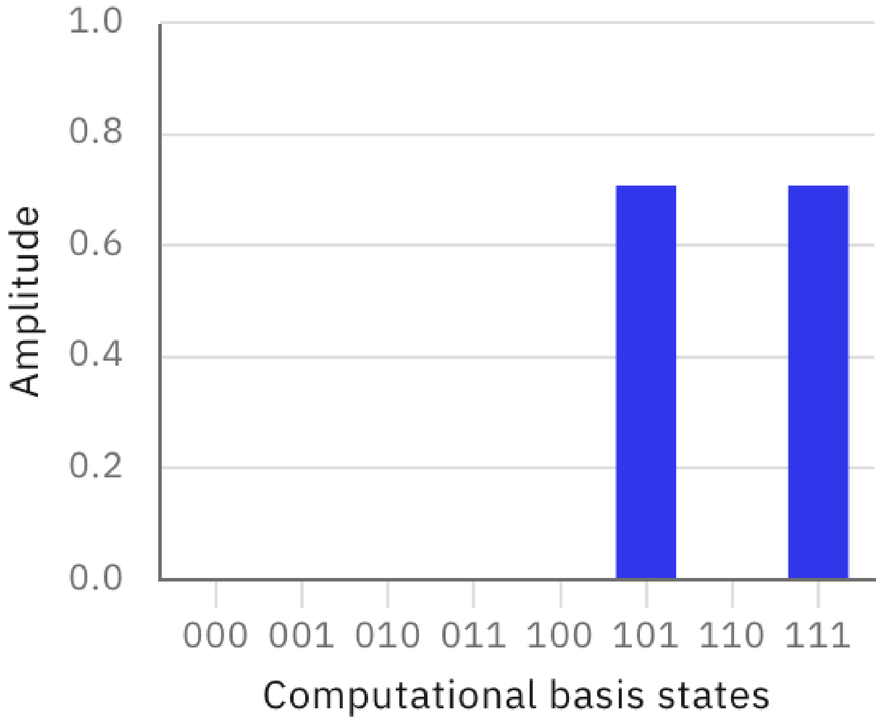
Basis encoding encodes a classical P-bit string into a computational basis (Z-basis) state of a P-qubit system.

Data set

$$X = \{x_1 = 101, \quad x_2 = 111\}$$

Quantum State

$$|x\rangle = \frac{1}{\sqrt{2}}(|101\rangle + |111\rangle)$$



Amplitude encoding

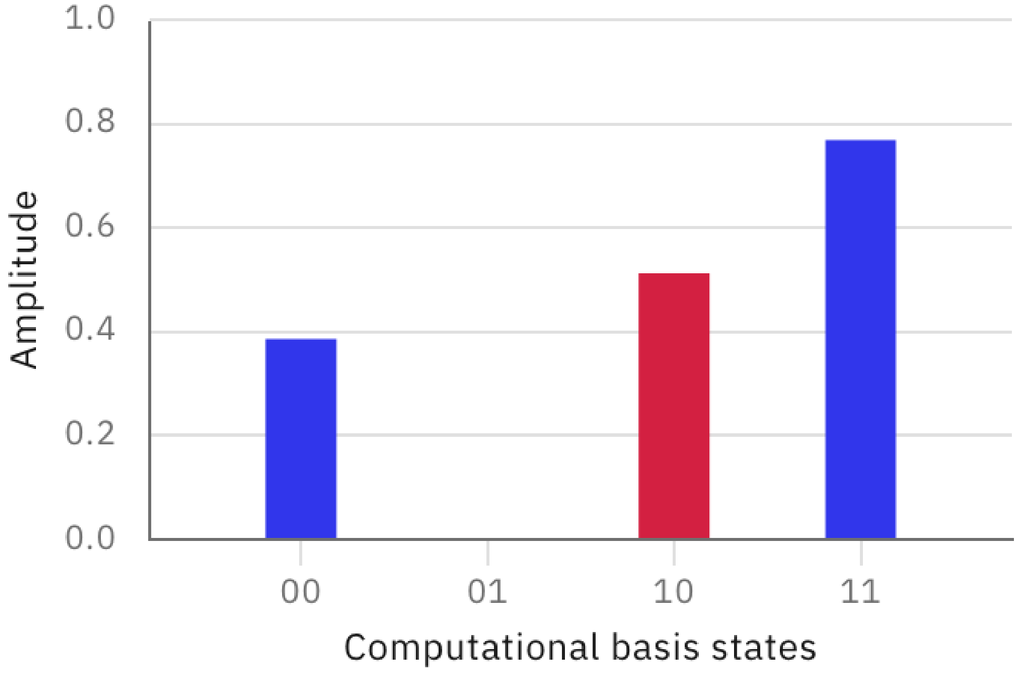
Amplitude encoding encodes data into the amplitudes of a quantum state.
If the computational basis is redundant, set its amplitude to zero.

Data set

$$X = \left\{ x_1 = (1.5, 0), \quad x_2 = (-2, 3) \right\}$$

Quantum State

$$|x\rangle = \frac{1}{\sqrt{15.25}} (1.5|00\rangle - 2|10\rangle + 3|11\rangle)$$



Angle encoding

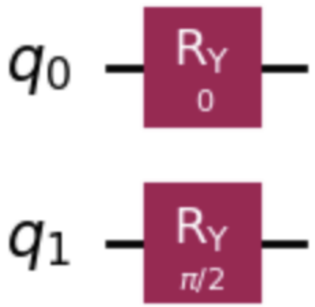
Angle encoding encodes data to the angle of a rotating gate, such as RX gate or RY gate.

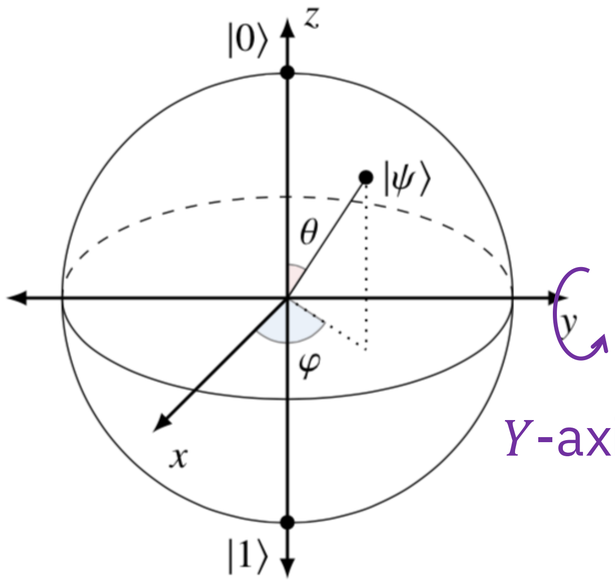
Data point

$$x = (x_1, x_2)$$

Quantum State

$$S_x = RY(x_1) \otimes RY(x_2)$$





Y-axis rotation

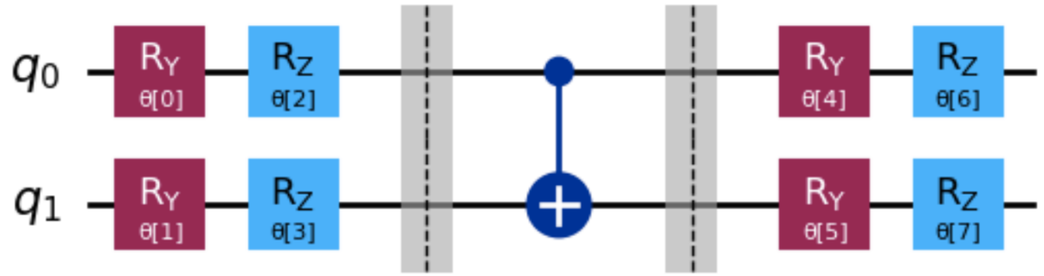
Enhanced angle encoding

Enhanced angle coding encodes data to the phase. It also includes arbitrary unitary operations (e.g., CNOT).
It is also possible to encode multiple data into a single qubit.

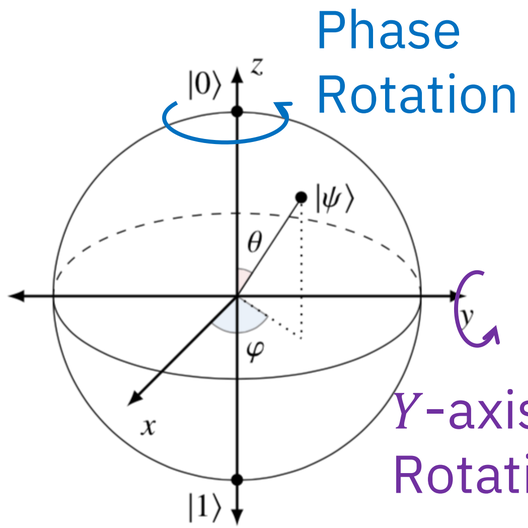
$x = (x_1, \quad x_2)$ ➡ $S_x = P(x_2)RY(x_1)$



$x = (x_1, \quad \cdots, x_8)$ ➡



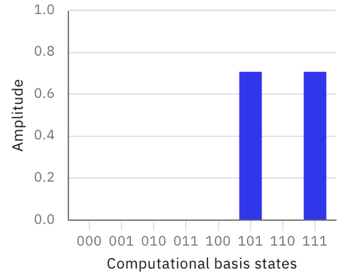
Encode 8 data with just 2 qubits



Typical Data encoding methods for QML

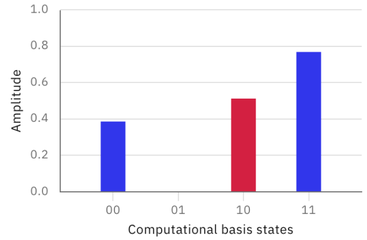
1. Basis encoding

$$X = \{x_1 = 101, \quad x_2 = 111\} \Rightarrow |x\rangle = \frac{1}{\sqrt{2}}(|101\rangle + |111\rangle)$$



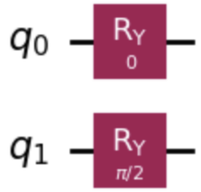
2. Amplitude encoding

$$X = \{x_1 = (1.5, 0), \quad x_2 = (-2, 3)\} \Rightarrow |x\rangle = \frac{1}{\sqrt{15.25}}(1.5|00\rangle - 2|10\rangle + 3|11\rangle)$$



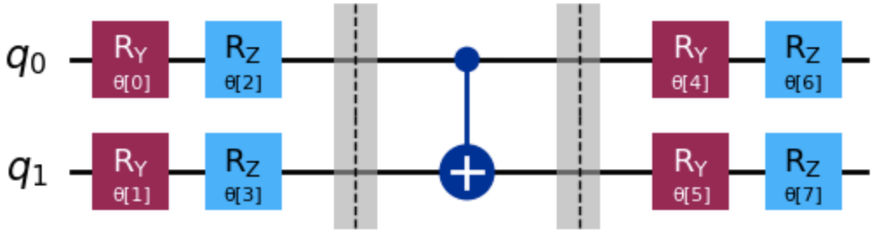
3. Angle encoding

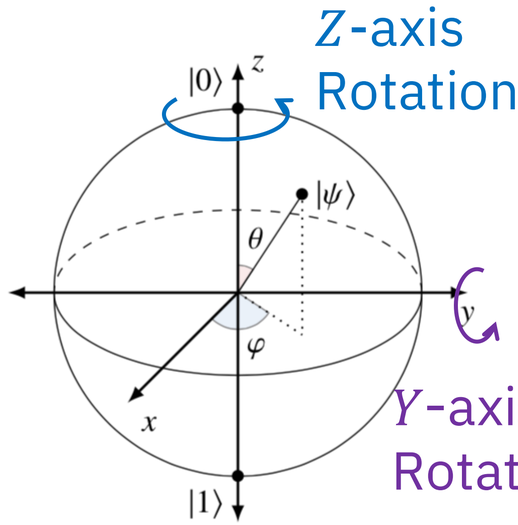
$$x = (x_1, \quad x_2) \Rightarrow S_x = RY(x_1) \otimes RY(x_2)$$



4. Enhanced angle encoding

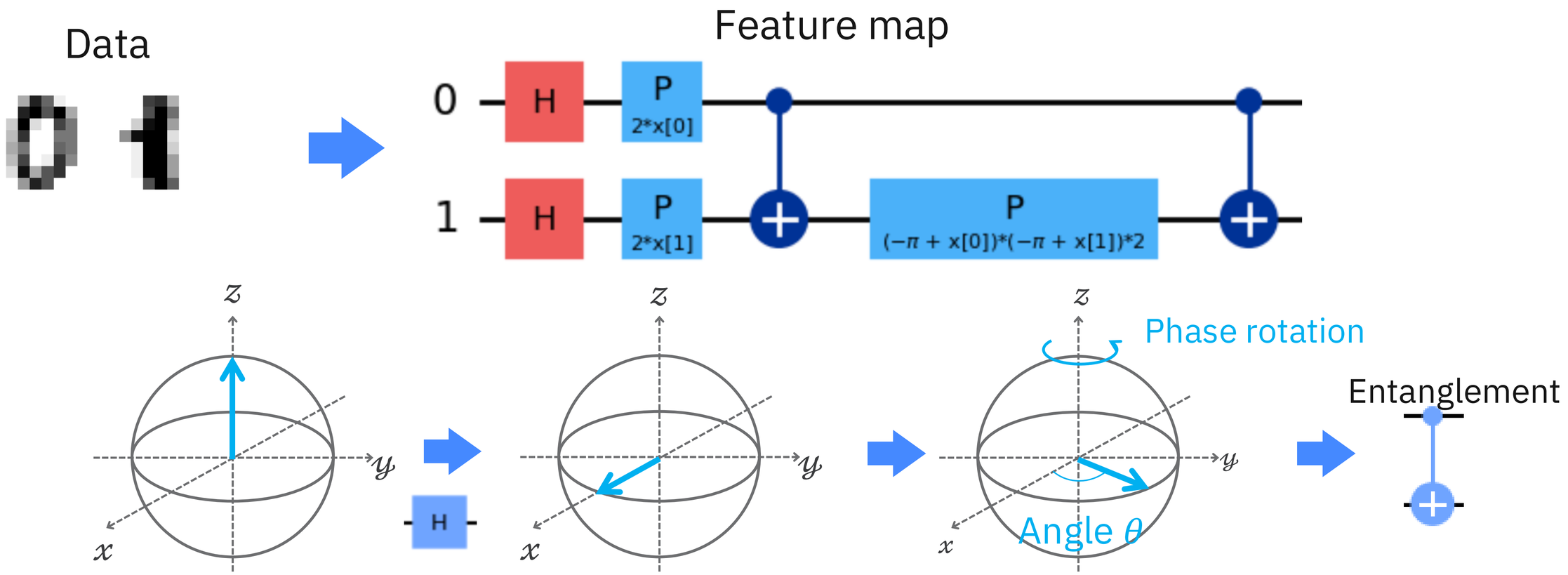
$$x = (x_1, \quad x_2) \Rightarrow$$





Qiskit built-in encoding scheme

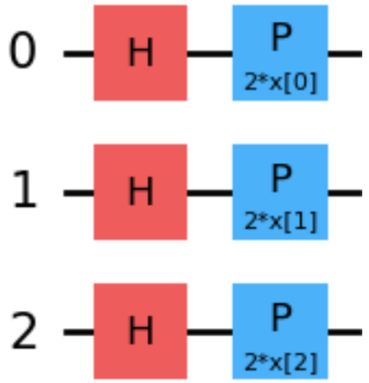
“Feature Map” is an enhanced angle encoding scheme.



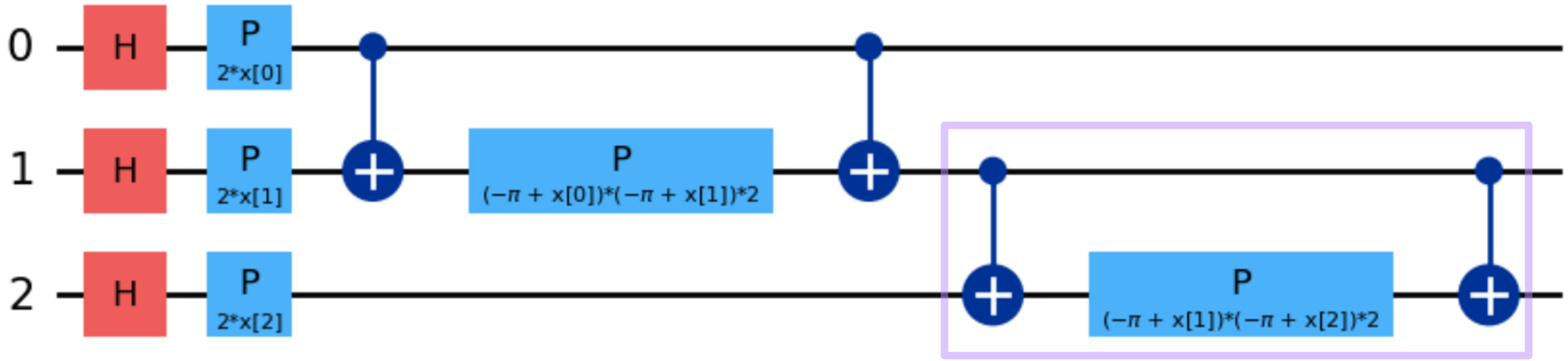
Quantum feature map

In case of 3-dimension data

Z feature map

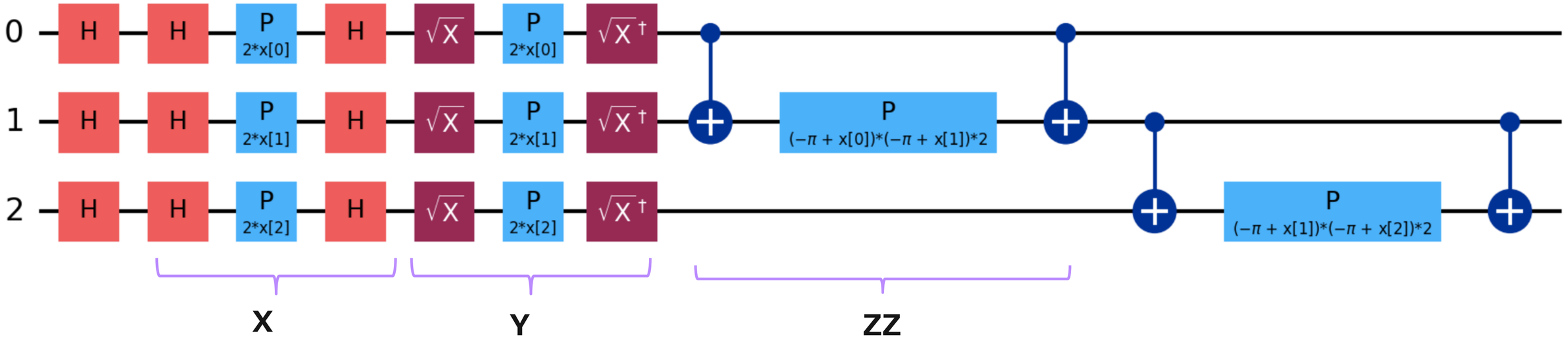


ZZ feature map



$R_{ZZ}(\theta)$

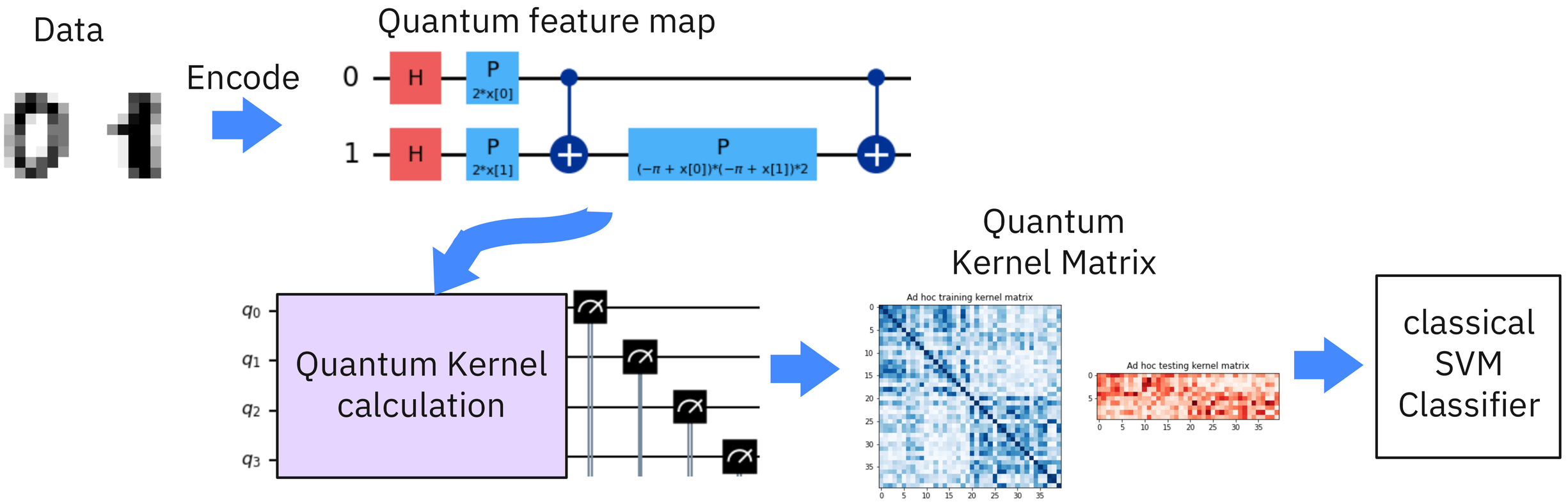
Pauli feature map (in case of using X, Y, and ZZ)



Equivalent circuit $X = HZH, \quad Y = R_X(\pi/2)Z\,R_X(-\pi/2)$

Quantum Kernel SVM

1. Encode the data into quantum state by a quantum feature map.
2. Calculate Quantum kernel using quantum computer.
3. Learn and classify the data with classical SVM using classical computer.



Quantum kernel is inner product

The kernel function takes vectors in the feature-mapped space as arguments and returns their inner product, i.e. $K(x, y) = \langle \Phi(x) | \Phi(y) \rangle$.

- The data vector is: $\overrightarrow{x_i}$
- The circuit responsible for encoding and mapping $\overrightarrow{x_i}$ is: $\Phi(\overrightarrow{x_i})$

Then, we can denote the mapped states:

$$|\psi(\overrightarrow{x_i})\rangle = \Phi(\overrightarrow{x_i})|0\rangle^{\otimes N}$$
$$|\psi(\overrightarrow{x_j})\rangle = \Phi(\overrightarrow{x_j})|0\rangle^{\otimes N}$$

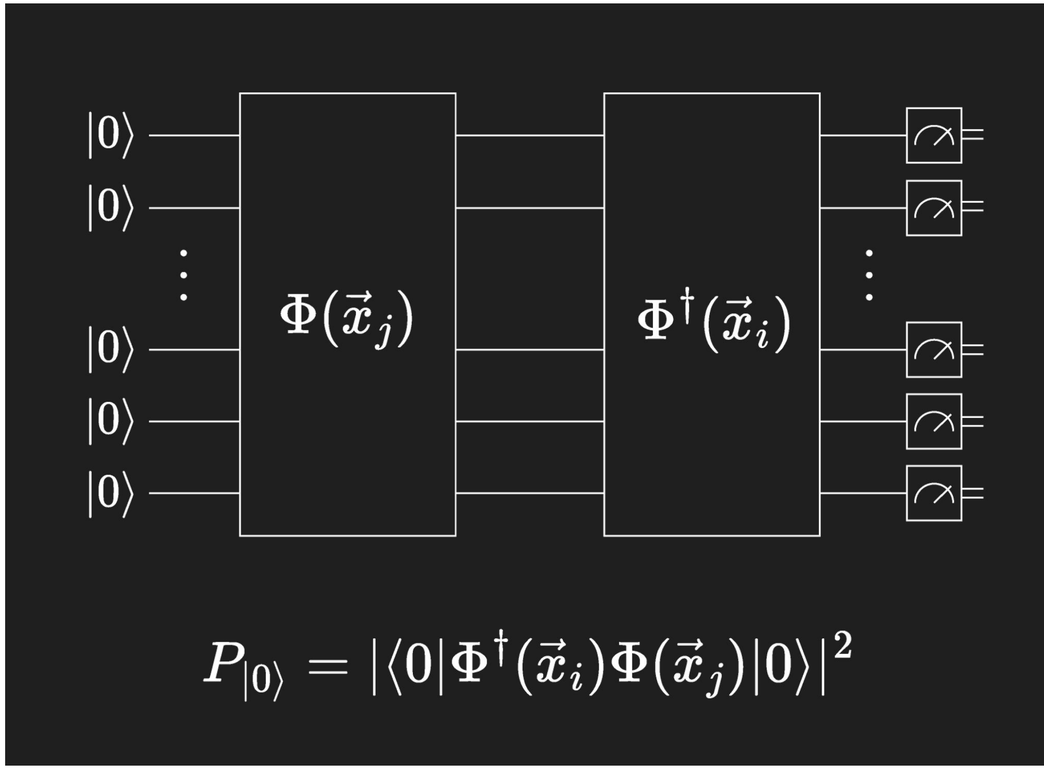
Our desired kernel entry is the inner product is:

$$\langle \psi(\overrightarrow{x_j}) | \psi(\overrightarrow{x_i}) \rangle = \langle 0|^{\otimes N} \Phi^{\dagger}(\overrightarrow{x_j}) \Phi(\overrightarrow{x_i}) |0\rangle^{\otimes N}$$

The probability of then measuring the state $|0\rangle^{\otimes N}$ is:

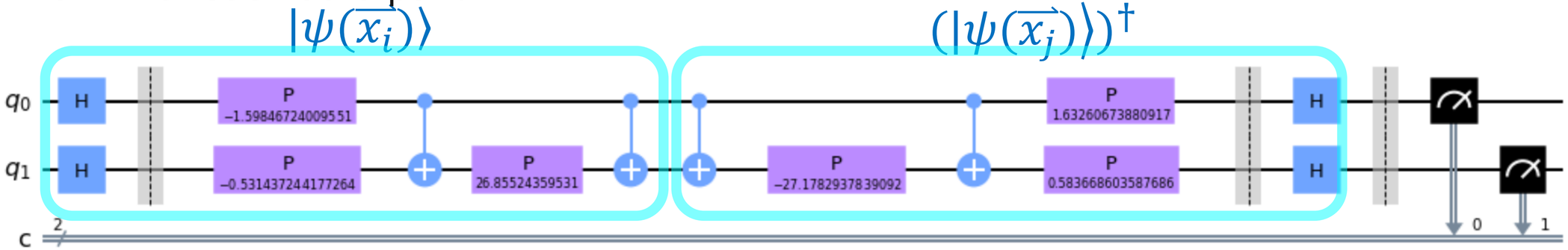
$$P_0 = \left| \langle 0|^{\otimes N} \Phi^{\dagger}(\overrightarrow{x_j}) \Phi(\overrightarrow{x_i}) |0\rangle^{\otimes N} \right|^2$$

This is the value of the kernel entry.



Quantum kernel in detail

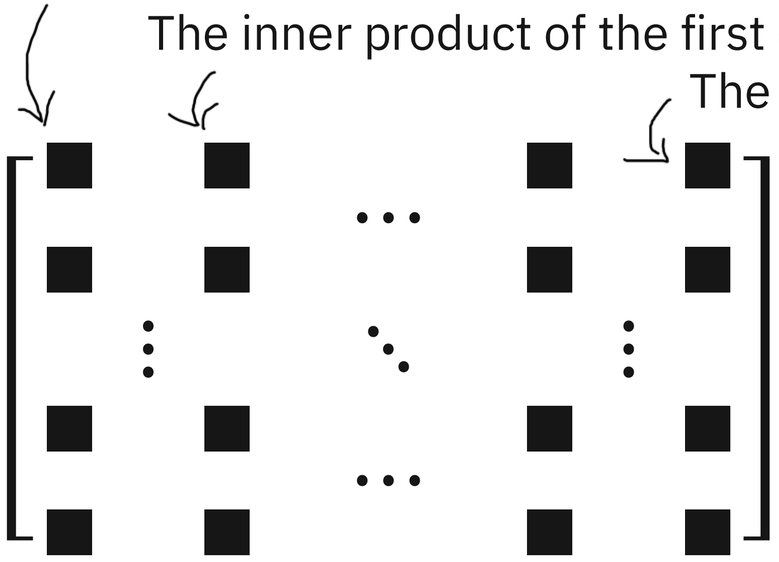
Calculate and measure the inner product, quantum kernel entry $\langle\psi(\vec{x_j})|\psi(\vec{x_i})\rangle$, to create a kernel matrix for each data pair.



The inner product of the first data and the first data

The inner product of the first data and the second data

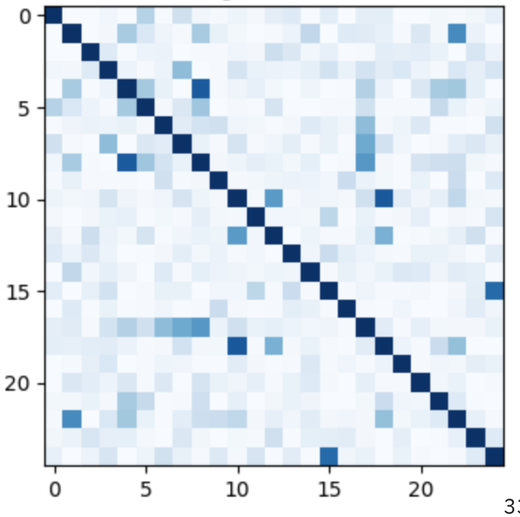
The inner product of the first data and the nth data



The inner product of the nth data and the nth data

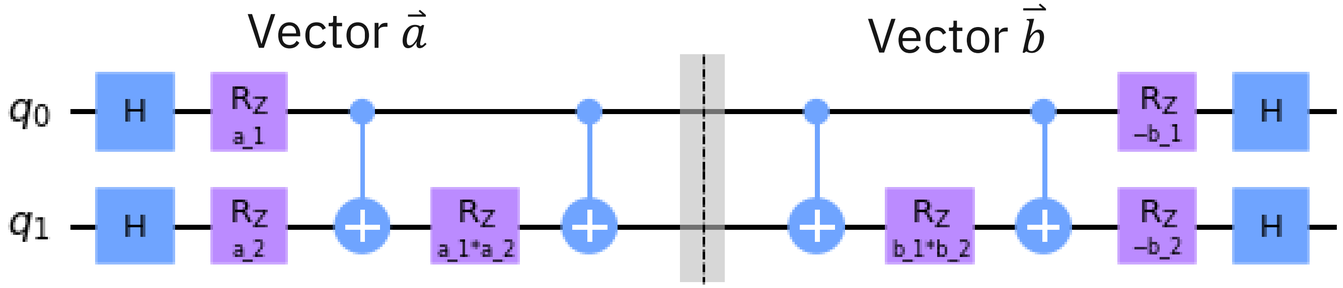
If we have 25 training data, then we will create 25 x 25 quantum kernel matrix.

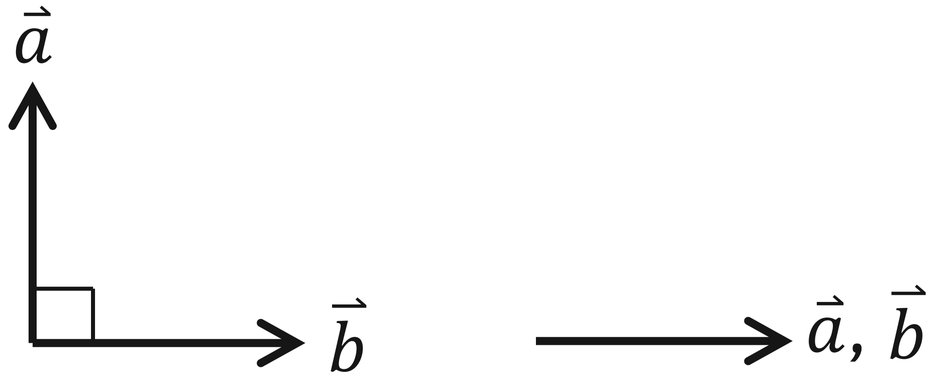
0: white
1: dark blue



Kernel calculation is a kind of similarity calculation

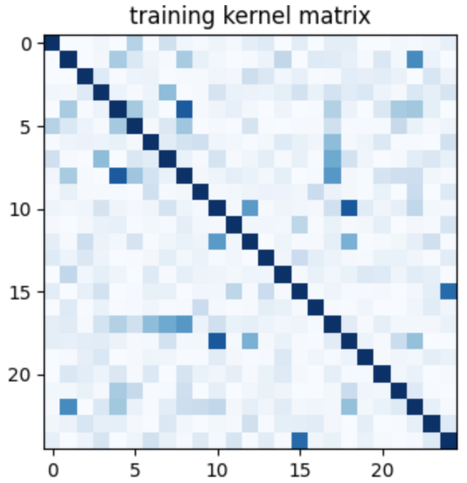
Kernel entry $\langle b|a\rangle$ is a fidelity of $|a\rangle$ and $|b\rangle$.





The inner product between orthogonal vectors is 0.

The inner product of two identical vectors is 1.



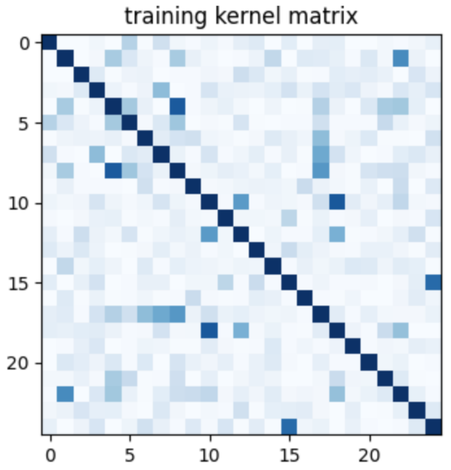
- Completely different feature: 0 (white)
- Same feature: 1 (dark blue or red)

How many calculations are needed for Quantum kernel

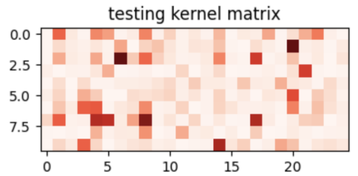
If you use 25 training data and 10 test data, we need to calculate all elements like:

- Kernel matrix: 25 x 25
- Test matrix: 10 x 25

25 x 25 matrix



10 x 25 matrix

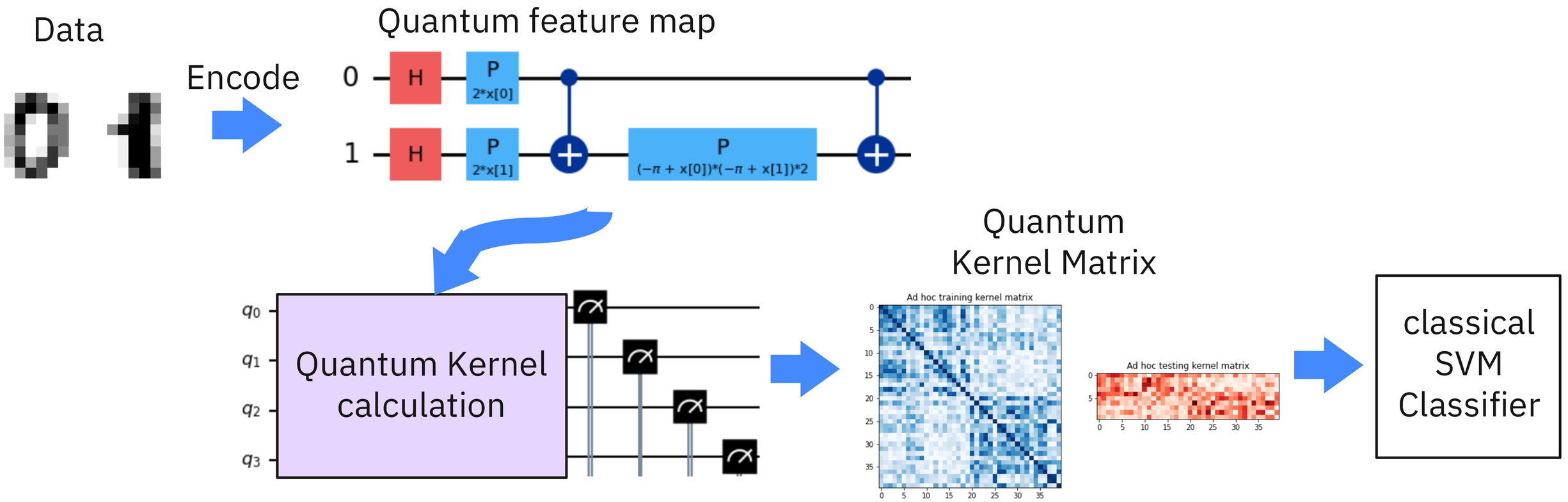


In the hands-on notebook, we will simplify the calculation of kernel matrix:

- 25 x 25 x ½ element for the symmetry to the diagonal
- Set 1 to the diagonal elements

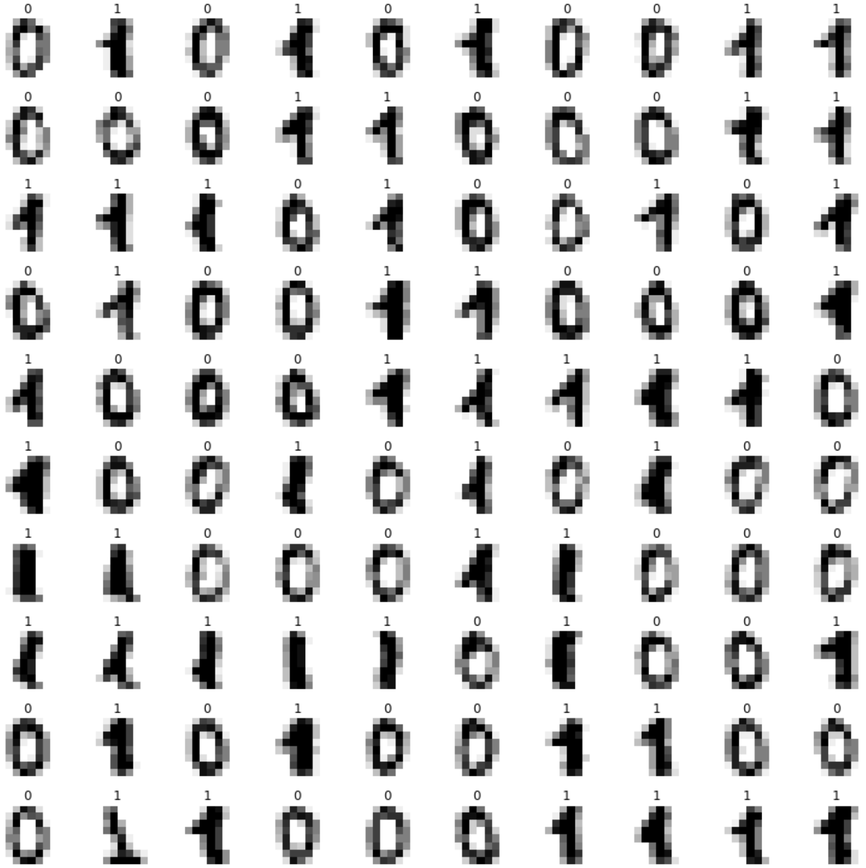
Quantum Kernel SVM

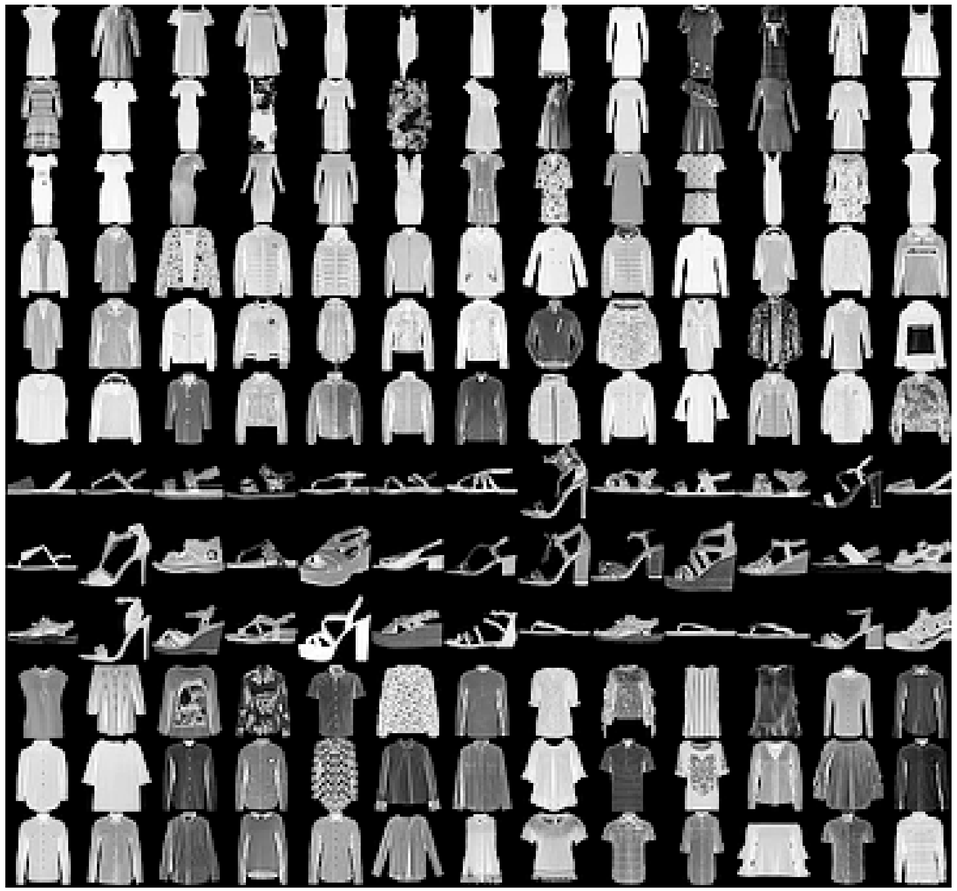
1. Encode the data into quantum state by a quantum feature map.
2. Calculate Quantum kernel using quantum computer.
3. Learn and classify the data with classical SVM using classical computer.



Hands-on

After learning classification handwritten digit number (MNIST), we will try to perform training classification fashion images.



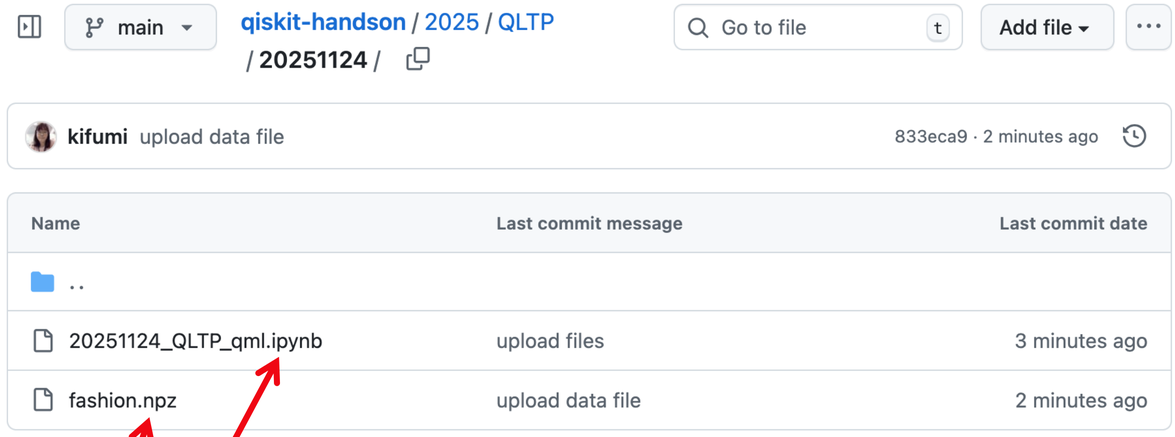


Qiskit hands-on

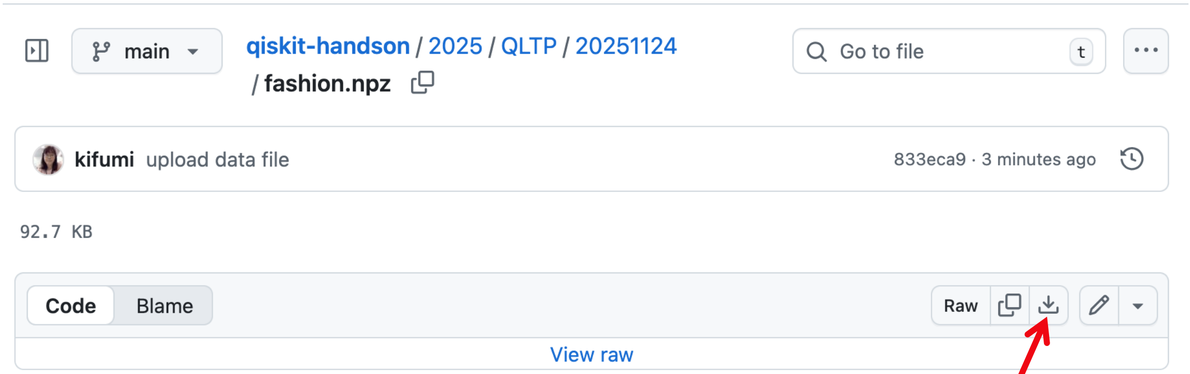
URL: ibm.biz/qltp251124

Download two files:

1. 20251124_QLTP_qml.ipynb
2. fashion.npz



1. Doble click the file one by one.



2. Download the file from this icon.

IBM **Quantum**

© 2025 International Business Machines Corporation